

Ismail Aissaoui
Email: i.aissaoui@esi-sba.dz
Phone: +213 660 70 77 96
Portfolio: ismail-aissaoui.vercel.app

Prof. Khalil Drira (LAAS-CNRS)
Dr. Christophe Chassot (LAAS-CNRS)
Dr. Thierry Monteko (LAAS-CNRS)

Subject: Application for PhD Position – Composing and configuring digital twin architectures (PC2)

Dear Prof. Drira, Dr. Chassot, and Dr. Monteko,

I am applying for the PC2 PhD position focused on "Composing and configuring digital twin architectures to meet user requirements" with great enthusiasm. With a strong engineering background in distributed systems, Quality of Service (QoS) optimization, and Edge-AI architectures, I am highly motivated to address the challenge of dynamically orchestrating digital twin components to satisfy strict performance, energy, and budgetary constraints.

In my work engineering the Industrial Gas Turbine Digital Twin, I faced the exact trade-offs your research seeks to optimize. Transporting high-frequency telemetry to the cloud for real-time anomaly detection incurred unacceptable latency and energy costs. To resolve this, I orchestrated an Edge-Optimized architecture, deploying lightweight Temporal Convolutional Networks (Edge-TCN) directly onto constrained edge nodes. By pushing computation to the edge, I dynamically balanced the system's QoS requirements (low latency inference) with resource constraints (limited bandwidth and power). This practical experience managing architectural trade-offs positions me perfectly to formalize these configuration strategies. I am eager to leverage my background in multi-objective optimization and Reinforcement Learning to automate the composition of Digital Twins, ensuring they can adapt their topology in real-time to meet shifting user Service Level Agreements (SLAs).

I am particularly excited to collaborate with the LAAS-CNRS teams to build these autonomic configuration managers. I am confident that my programming proficiency and deep understanding of distributed cyber-physical systems will allow me to contribute directly to the Artemis platform and the broader goals of the EDT program. Please find attached my resume and academic transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui